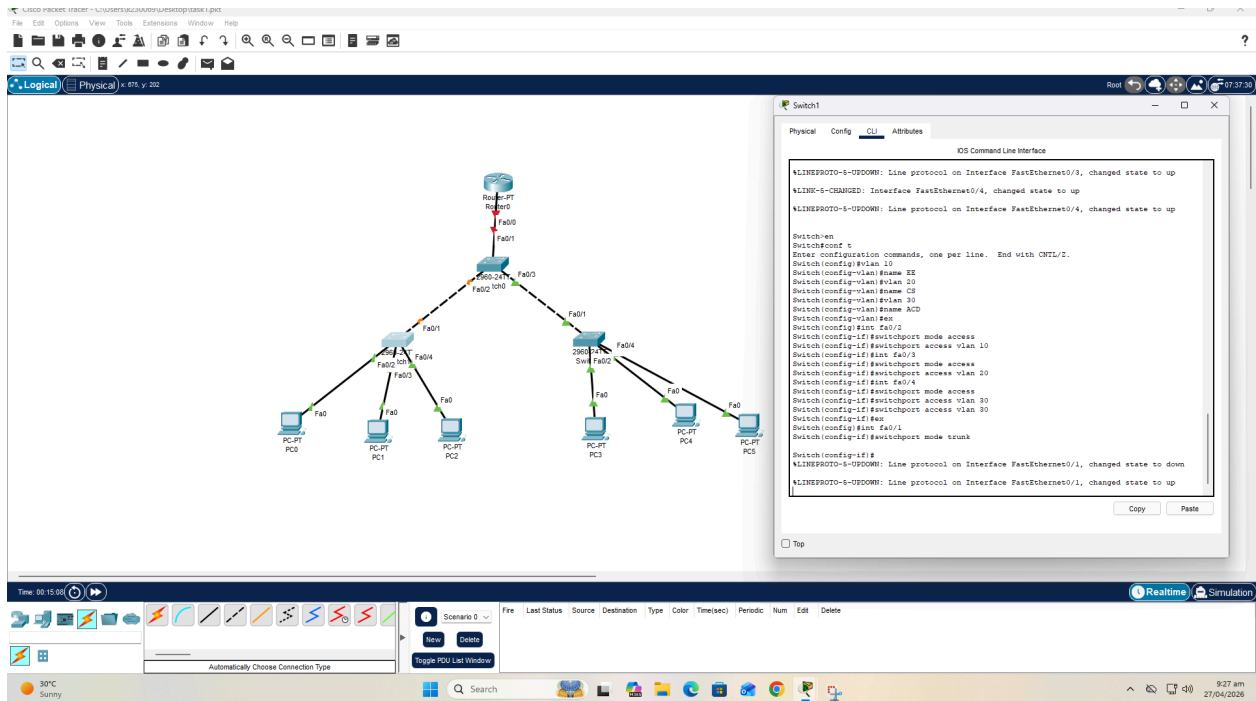


lab 12

task a



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name EE
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name CS
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name ACD
Switch(config-vlan)#ex
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Copy Paste

☐ Top

Cisco Packet Tracer - C:\Users\k230069\Desktop\task1.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical 44 y 8'

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```

%LINK-3-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-3-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name EE
Switch(config-vlan)#name 20
Switch(config-vlan)#name CS
Switch(config-vlan)#name 30
Switch(config-vlan)#name ACD
Switch(config-vlan)#ex
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINK-3-CHANGED: Interface FastEthernet0/1, changed state to down
%LINEPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
  
```

Copy Paste

Time 00:22:30

Realtime Simulation

Scenario 0

Now Delete

Topology List Window

File List Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

Automatically Choose Connection Type

29°C Sunny

Search

8:34 am 27/04/2026

 Switch2

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name EE
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name CS
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name ACD
Switch(config-vlan)#ex
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
|
```

Copy Paste

☐ Top

Switch0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port
type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/3 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/3 on VLAN0001. Inconsistent port
type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
|

Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport mode trunk
Switch(config-if)#ex
Switch(config)#vlan 10
Switch(config-vlan)#name EE
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name CS
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name ACD
Switch(config-vlan)#
Switch(config-vlan)#vlan 30
Switch(config-vlan)#exit
Switch(config)#switchport mode trunk
Switch(config)#
^
% Invalid input detected at '^' marker
```

Copy

Paste

☐ Top

Cisco Packet Tracer - C:\Users\k230069\Desktop\task1.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical = 45.1 y 102

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```

Router0#configure t
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#int fa0/0
Router0(config-if)#no shut

Router0(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router0(config-if)#int fa0/0.10
Router0(config-subif)#
Router0(config-subif)#int fa0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

Router0(config-subif)#encapsulation dot1q 10
Router0(config-subif)#ip add 192.168.1.1 255.255.255.0
Router0(config-subif)#int fa0/0.20
Router0(config-subif)#
Router0(config-subif)#int fa0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

Router0(config-subif)#int fa0/0.10
Router0(config-subif)#encapsulation dot1q 10
Router0(config-subif)#ip add 10.0.0.1 255.0.0.0
Router0(config-subif)#encapsulation dot1q 20
Router0(config-subif)#ip add 20.0.0.1 255.0.0.0
Router0(config-subif)#int fa0/0.30
Router0(config-subif)#
Router0(config-subif)#int fa0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up

Router0(config-subif)#encapsulation dot1q 30
Router0(config-subif)#

```

Copy Paste

Top

Time: 00:30:00

Scenario 0

File Log Status Source Destination Type Color Time(sec) Periodic Run Edit Delete

Now Delete

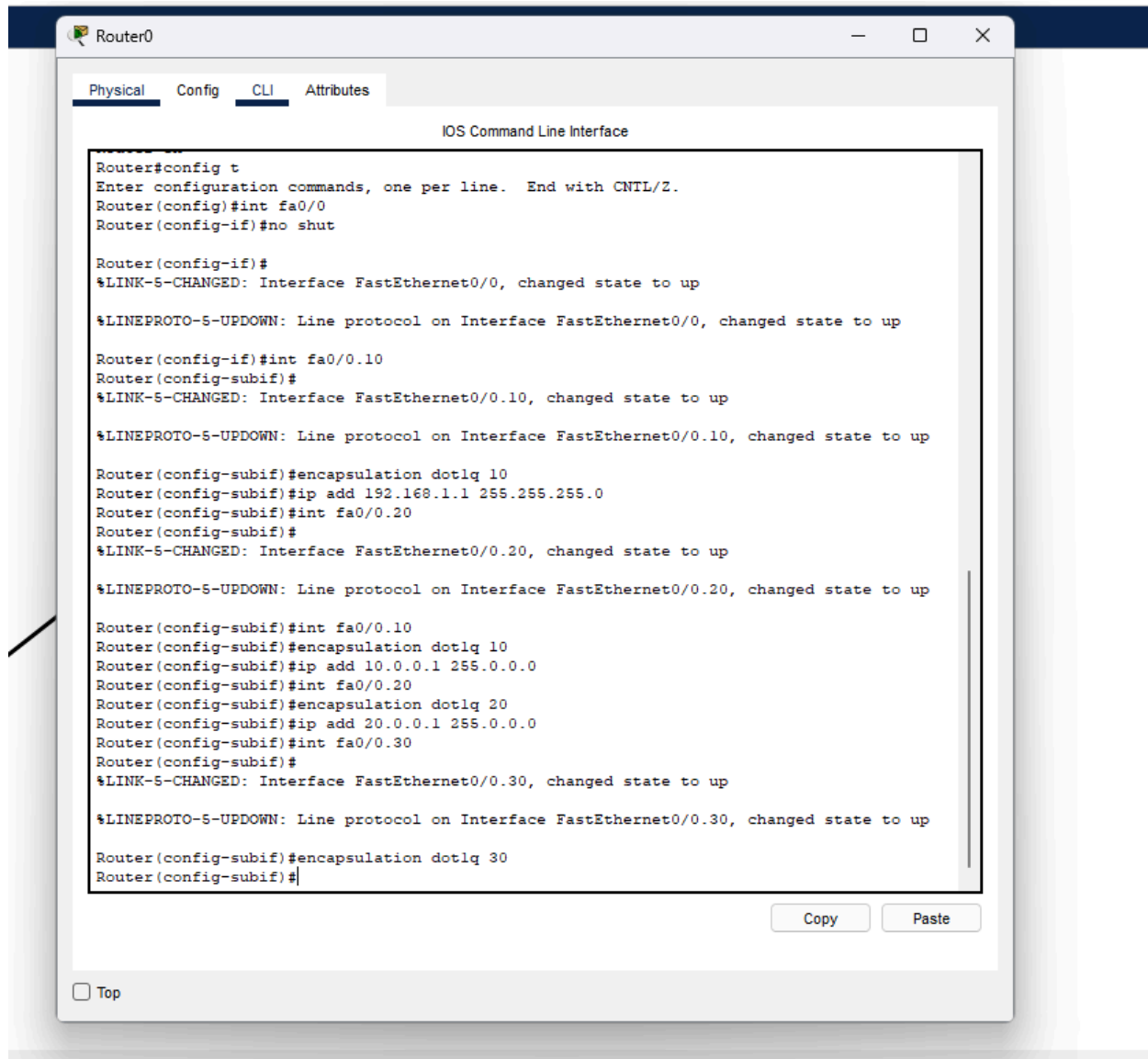
Toggle PCU List Window

Automatically Choose Connection Type

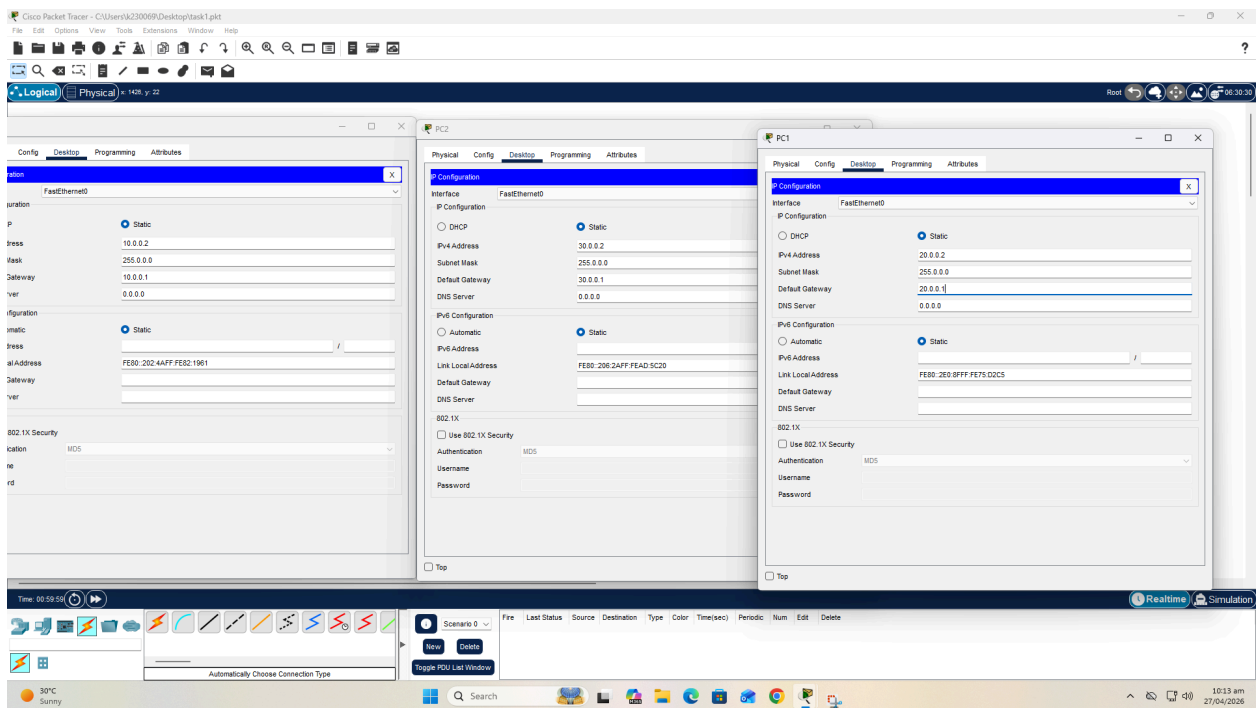
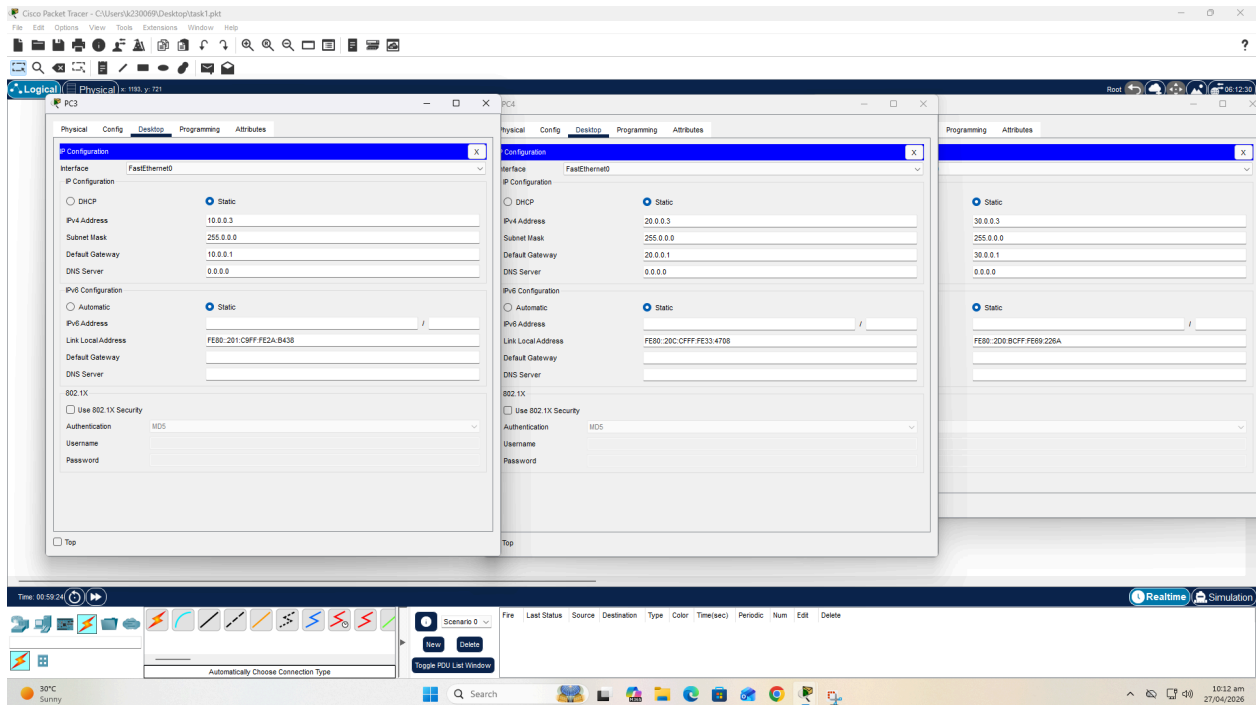
29°C Sunny

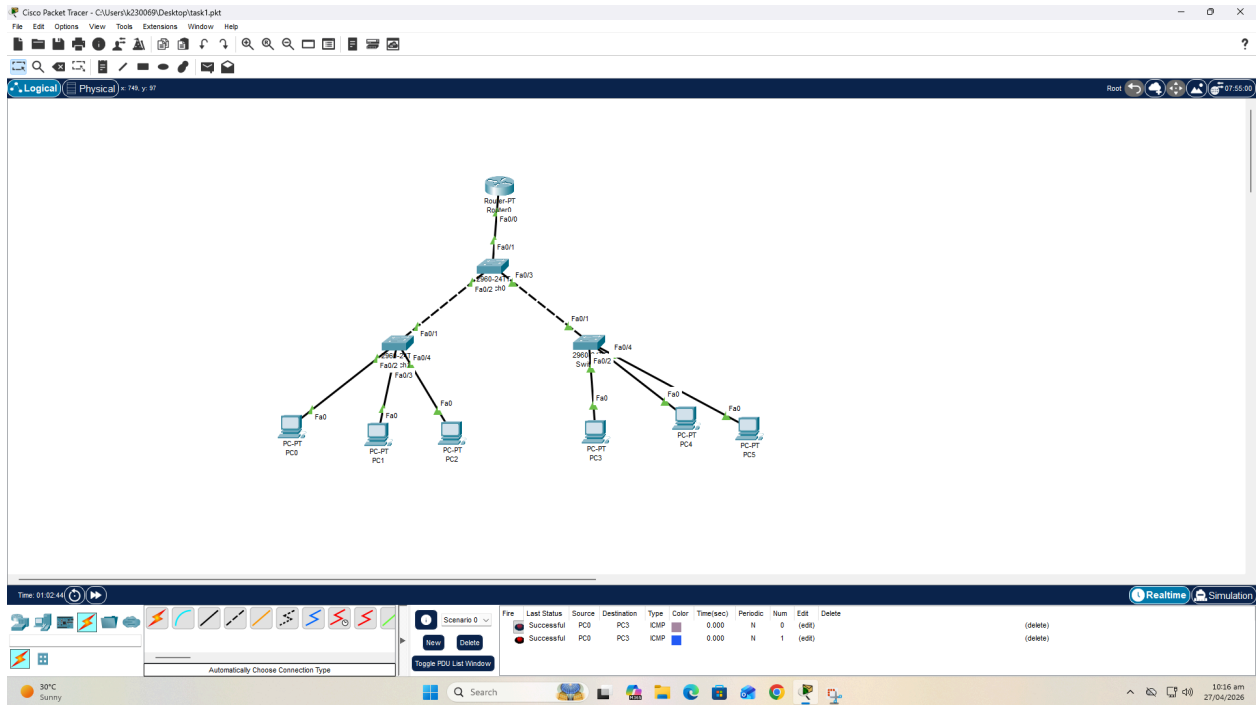
Search

8:51 am 27/04/2026

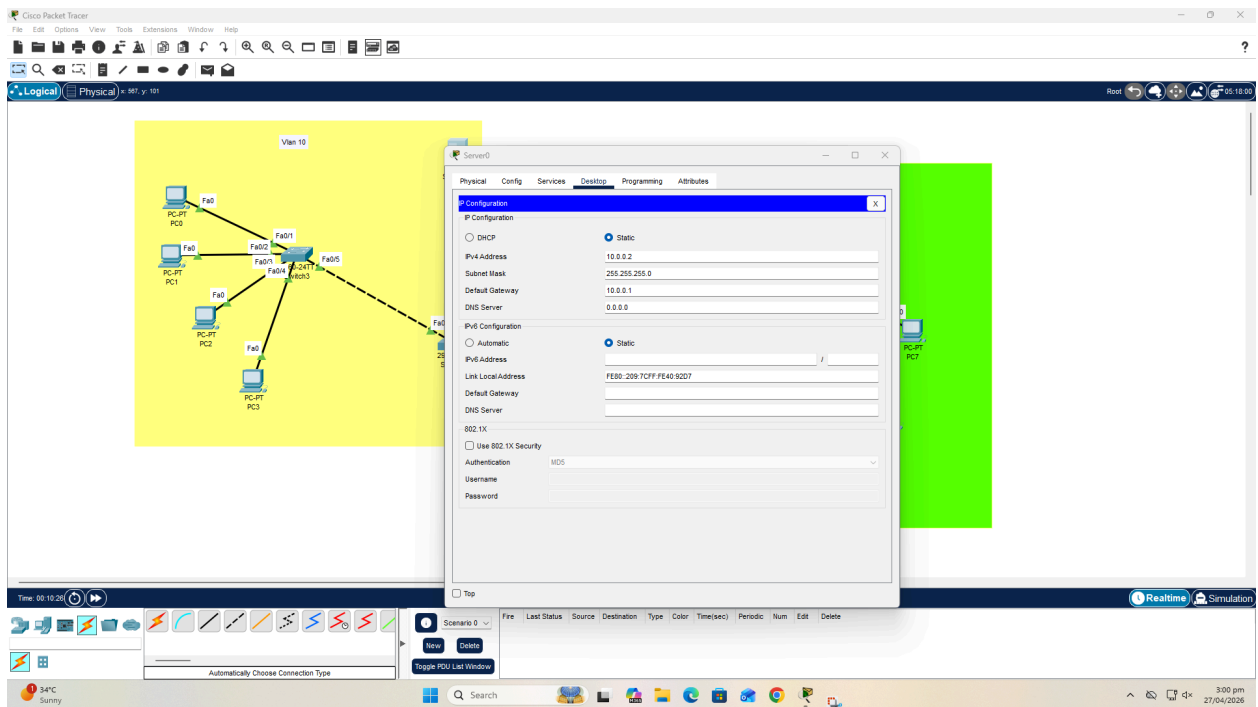


pcs ip





task b



Server0

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

vlan20

Default Gateway

192.168.20.1

DNS Server

0.0.0.0

Start IP Address :

192

168

20

10

Subnet Mask:

255

255

255

0

Maximum Number of Users :

246

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

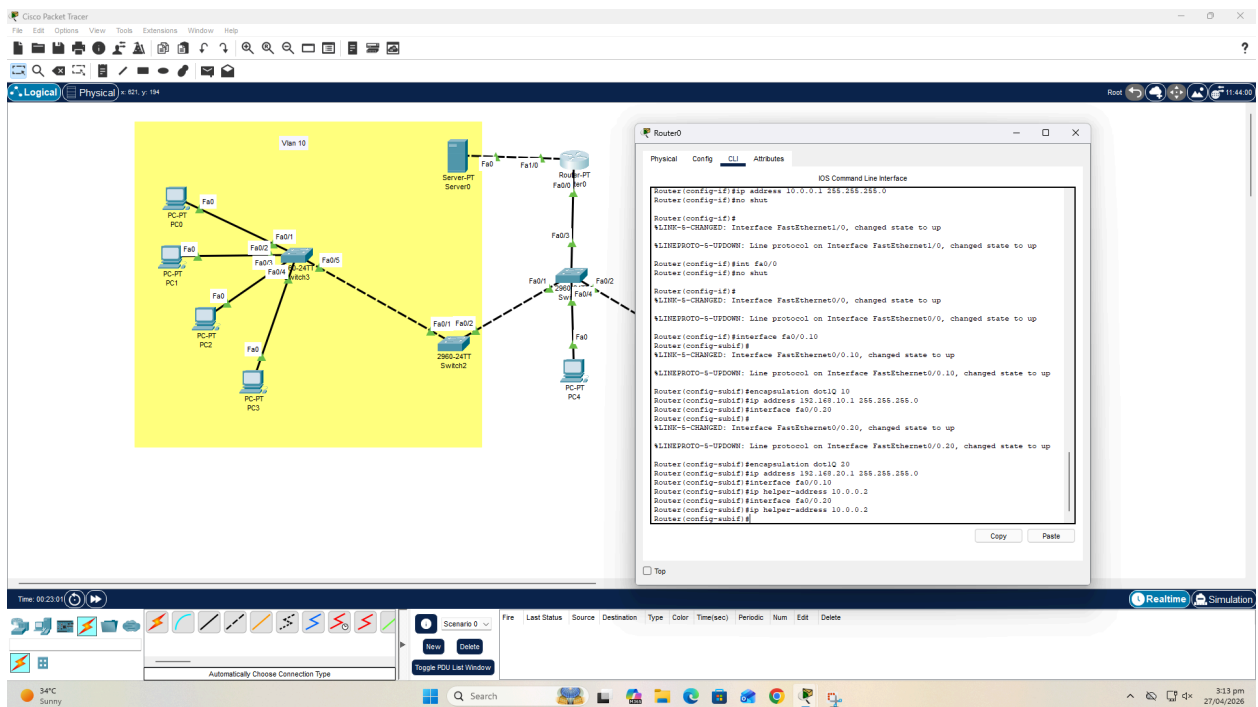
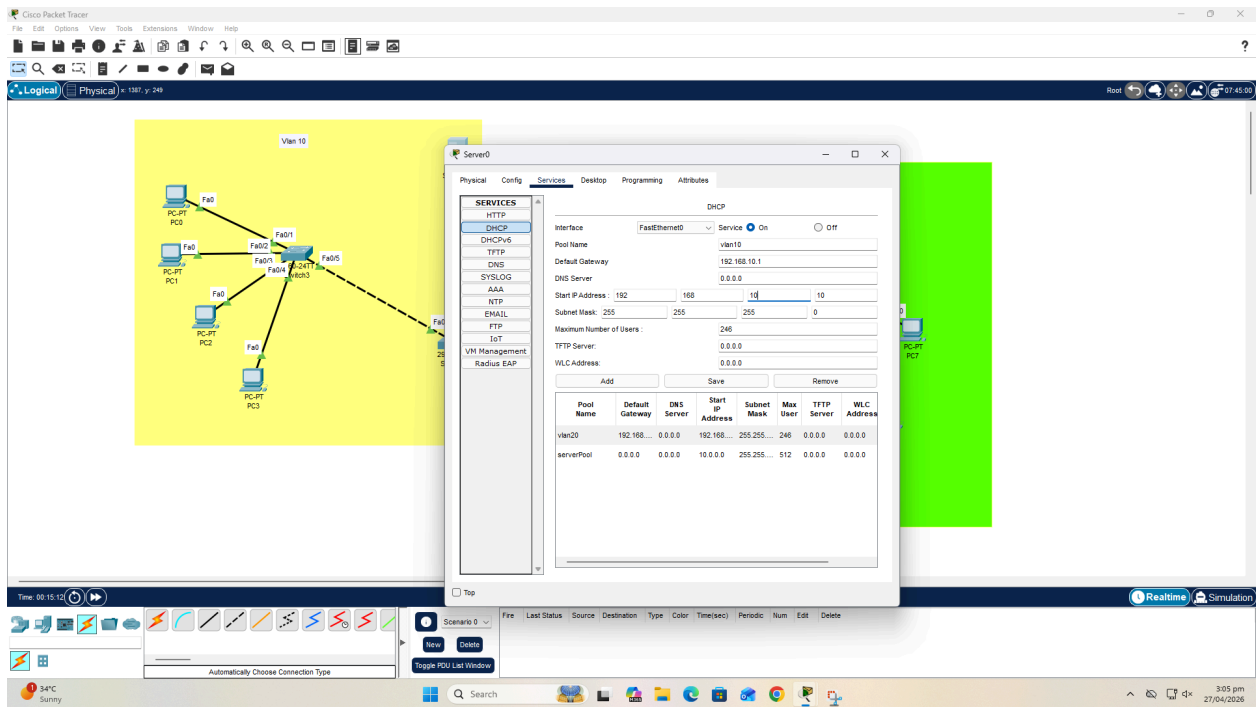
Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	10.0.0.0	255.255....	512	0.0.0.0	0.0.0.0

Top



Router0

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config-if)#int fa0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#interface fa0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface fa0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#interface fa0/0.10
Router(config-subif)#ip helper-address 10.0.0.2
Router(config-subif)#interface fa0/0.20
Router(config-subif)#ip helper-address 10.0.0.2
Router(config-subif)#
```

CopyPaste

☐ Top

Cisco Packet Tracer - C:\Users\catac\Downloads\lab12b.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 436, y: 9
Vlan 10
PC-PT PC1 Fa0
PC-PT PC2 Fa0
Switch0 Fa0/2 Fa0/3 Fa0/4 Fa0/5 Fa0/1

Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name a1
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name a2
Switch(config-vlan)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch FastEthernet0/1 (1).
Switch(config-if)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch(config-if)#int fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Root
04:18:00

Cisco Packet Tracer - C:\Users\catac\Downloads\lab12b.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 684, y: 378
Vlan 10
PC-PT PC1 Fa0
PC-PT PC2 Fa0
Switch0 Fa0/2 Fa0/3 Fa0/4 Fa0/5 Fa0/1

Switch2
Physical Config CLI Attributes
IOS Command Line Interface
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/2 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch(config-if)#int fa0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#

Root
05:32:30

Realtime Simulation
Color Time(sec) Periodic Num Edit Dt
0.000 N 0 (e... (d
0.000 N 1 (e... (d
0.000 N 2 (e... (d

Cisco Packet Tracer - C:\Users\catac\Downloads\lab12b.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x:788, y:315
Time: 00:24:29

Switch4
Physical Config CLI Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ai
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name cs
Switch(config-vlan)#int range fa0/2-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#int fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Copy Paste

Realtime Simulation
Color Time(sec) Periodic Num Edit Dt
0.000 N 0 (e... (d
0.000 N 1 (e... (d
0.000 N 2 (e... (d

PC0

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name

PC0

Interfaces

FastEthernet0

Gateway/DNS IPv4

☒ DHCP
☐ Static

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

Gateway/DNS IPv6

☐ Automatic
☒ Static

Default Gateway

DNS Server

	Successful	Server0	PC0	ICMP		0.000	N	6	(edi...
	Successful	PC2	Server0	ICMP		0.000	N	7	(edi...
	Successful	PC6	PC2	ICMP		0.000	N	8	(edi...

lab 12

17